24AI202 — Unit IV: Machine Learning

Regression

in Machine Learning

Predicting Numbers — Types • Algorithms • Evaluation • Applications

• R.M.K. Engineering College
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Slides cover

Regression — Predicting a Continuous Number

Regression is a Supervised Learning task where the model learns from labeled data to predict a CONTINUOUS NUMERIC VALUE — any number on a scale — for new inputs. The output is a quantity, not a category.

Input Features (X) + Correct Output Numbers (Y) → Trained Model → Predict $\hat{Y}$ (number)

House Price

How much does this house cost?

Inputs: Area, rooms, location, floor

₹52,00,000

Exam Score

How many marks will this student score?

Inputs: Study hours, attendance, sleep

78 / 100

Rainfall

How much will it rain tomorrow?

Inputs: Humidity, pressure, wind speed

142 mm

Temperature

What will the temperature be today?

Inputs: Month, location, pressure, season

34.5 °C

Regression vs Classification — Key Differences

Aspect	Regression	Classification
Output type	Continuous number	Discrete class label
Question asked	How much? How many? How long?	Which category? Which class?
Example output	₹52L, 78 marks, 142mm, 34.5°C	Spam, Diabetic, Mango, Pass
Algorithms	Linear, Ridge, Lasso, SVR, DT Reg	LR, DT, RF, SVM, KNN, NB
Evaluation metrics	MAE, MSE, RMSE, R^2 Score	Accuracy, Precision, Recall, F1
Loss function	Mean Squared Error (MSE)	Cross-Entropy Loss
Real-world eg	House price, Rainfall, Stock	Fraud, Disease, Sentiment

Memory rule: Answer is a NUMBER → Regression | Answer is a WORD → Classification

From Raw Data to Predicted Number

1 **Collect labeled data**

Gather dataset where each sample has input features + correct numeric output (price, marks, rainfall)

2 **Feature engineering**

Select relevant features, create new ones, handle missing values, normalise or scale numbers

3 **Split: Train / Test**

80% of data → Train (model learns). 20% → Test (evaluate on unseen data to check generalisation)

4 **Choose algorithm**

Linear, Polynomial, Ridge, Lasso, SVR — based on whether the relationship is linear or non-linear

5 **Train the model**

Algorithm fits a curve/line through data by minimising Mean Squared Error (MSE loss function)

6 **Evaluate on test set**

Compare predicted numbers vs actual numbers using MAE, RMSE, R^2 Score on the 20% test data

7 **Tune hyperparameters**

Adjust regularisation strength (α), polynomial degree, kernel

8 **Predict new values**

Feed new unseen input → model outputs a precise numeric

Regression minimises the error between predicted number and actual number — using MSE as the loss function

Linear Regression — Best-Fit Straight Line

Fits the best straight line through data points by minimising the sum of squared differences between actual values and predicted values (called Ordinary Least Squares).

Formula

Simple: $\hat{Y} = mX + c$

m = slope (how steep the line) c = intercept (where line crosses Y-axis)

Multiple: $\hat{Y} = w_1x_1 + w_2x_2 + \ldots + w_nx_n + b$

w = weights (learned by algorithm) b = bias

Worked example

Predict house price from area:

$\hat{Y} = 5000 \times \text{Area} + 2{,}00{,}000$

Area = 1000 sq.ft

$\hat{Y} = 5000 \times 1000 + 2{,}00{,}000 = ₹52{,}00{,}000$

Predicted house price → ₹52,00,000

Assumptions

Linear relationship, normally distributed errors, no multicollinearity

Use when

Simple or multiple linear relationship between features and output

Limitation

Cannot model curves — fails for non-linear patterns in data

Metrics

Evaluate using MAE, RMSE, and R^2 Score on test data

Polynomial Regression — Fitting Curves

When the relationship between input and output is CURVED (non-linear), Polynomial Regression fits a curved line by adding powers of the input features.

Formula (degree 2 — Quadratic)

$$\hat{Y} = w_0 + w_1X + w_2X^2$$

Degree 3: $\hat{Y} = w_0 + w_1X + w_2X^2 + w_3X^3$ (cubic)

Degree 1 (Linear)

Shape: Straight line

Use: Linear relationships — house price vs area

Overfit risk: Low

Degree 2 (Quadratic)

Shape: Parabola (U-curve)

Use: Diminishing returns — speed vs stopping distance

Overfit risk: Low

Degree 3+ (High deg)

Shape: S-curve / wavy line

Use: Complex patterns — use with caution

Overfit risk: HIGH

Key rule: Higher degree → more flexible but more prone to overfitting. Always validate with test data.

Ridge & Lasso — Regularised Regression

Both Ridge and Lasso are LINEAR REGRESSION + a PENALTY TERM that prevents overfitting by shrinking model weights. This is called Regularisation.

Ridge Regression (L2)

$$\text{Loss} = \text{MSE} + \alpha \times \Sigma w_i^2$$

Penalty: sum of SQUARES of weights (L2 norm)

Effect:	Shrinks all weights towards zero — none become exactly 0
Best for:	Data with many correlated features (multicollinearity)
α (alpha):	Controls penalty strength. $\alpha=0 \rightarrow$ plain linear. $\alpha\rightarrow\infty \rightarrow$ all weights$\rightarrow 0$
Keeps:	ALL features in the model — none removed

Lasso Regression (L1)

$$\text{Loss} = \text{MSE} + \alpha \times \Sigma|w_i|$$

Penalty: sum of ABSOLUTE values of weights (L1 norm)

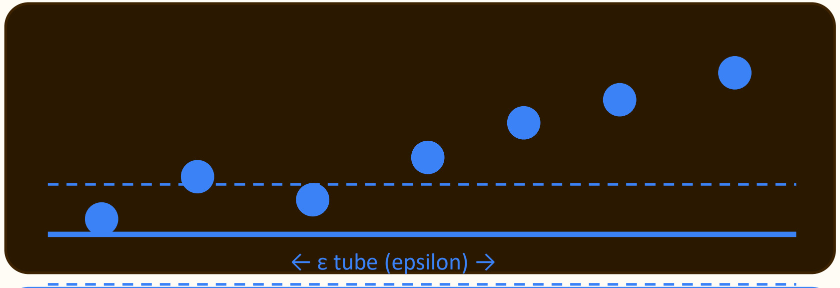
Effect:	Can shrink weights to EXACTLY zero — removes features
Best for:	High-dimensional data with many irrelevant features
Feature sel.:	Performs automatic feature selection — unique to Lasso
Keeps:	Only the most important features — sparse model

Ridge: keeps all features, shrinks weights | Lasso: removes irrelevant features, sparse model | Both prevent overfitting

SVR and Decision Tree Regression

SVR — Support Vector Regression

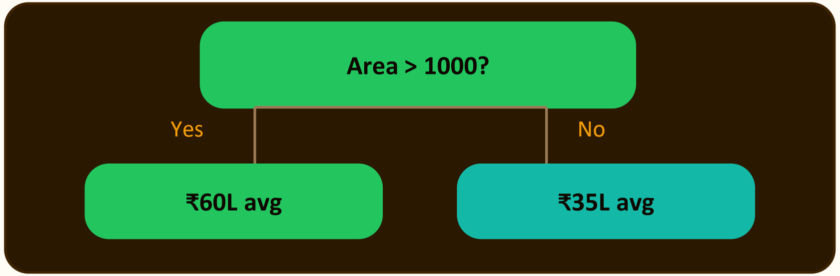
Finds a tube (epsilon-insensitive zone) around the data. Only errors OUTSIDE the tube are penalised. Points on the boundary are support vectors.



Kernel trick:	RBF kernel handles non-linear patterns — like SVM
ε (epsilon):	Width of tube. Points inside = no penalty
Use when:	Complex non-linear data, handles outliers well
Advantage:	Robust to outliers — only boundary points matter

Decision Tree Regression

Uses the same tree-splitting mechanism as classification DT but predicts a NUMBER (average of training samples in each leaf node) instead of a class label.



Splitting:	Uses variance reduction instead of Gini impurity
Leaf value:	Average of all training samples in that leaf
Advantage:	Handles non-linear data without feature scaling
Use when:	Non-linear data; want interpretable regression model

All Regression Algorithms

Algorithm	How it works	Best for	Interpretable	Speed
Linear Regression	Best-fit straight line (OLS)	Linear relationships	Yes	Very fast
Polynomial Regression	Curved line — adds X^2, X^3 terms	Non-linear curved data	Yes	Fast
Ridge Regression	Linear + L2 penalty (Σw_i^2)	Correlated / many features	Yes	Fast
Lasso Regression	Linear + L1 penalty ($\Sigma\|w_i\|$)	Feature selection needed	Yes	Fast
SVR	Tube around data; outlier robust	Non-linear, outlier data	No	Medium
DT Regression	Tree splits → leaf avg prediction	Non-linear, no scaling needed	Yes	Fast
Neural Network	Multi-layer nonlinear transformation	Complex patterns, big data	No	Slow

Start with Linear Regression → if non-linear add Polynomial → use Ridge/Lasso if overfitting occurs

How We Measure Regression Performance

We cannot use Accuracy for regression (output is a number, not a class). We measure how CLOSE the predicted numbers are to the actual values using error-based metrics.

Metric	Name / Formula	Description	Goal
MAE	**Mean Absolute Error** $(1/n) \times \Sigma\ \lvert actual - predicted \rvert$	Average error in the SAME UNIT as output. Easy to interpret. *House price: avg ₹2 lakh error in predictions*	Lower is better
MSE	**Mean Squared Error** $(1/n) \times \Sigma\ (actual - predicted)^2$	Squares errors — PENALISES large errors more. Sensitive to outliers. *Used as the loss function during model training*	Lower is better
RMSE	**Root Mean Squared Error** $\sqrt{MSE}$	Square root of MSE — brings back to original units. Most popular. *Rainfall: typical prediction error is 12.4 mm (RMSE)*	Lower is better
R^2	**R-Squared (Coefficient of Determination)**	% of variance explained by model. $R^2=1$ → perfect. $R^2=0$ → no better than mean.	Closer to 1.0 is better

MAE/MSE/RMSE: Lower = Better | R^2: Closer to 1.0 = Better | Perfect model: MAE=0, R^2=1.0

Overfitting vs Underfitting — The Core Challenge

Underfitting (High Bias)

Train error:	High
Test error:	High
Degree:	Too low (deg 1)

Cause: Model too simple — cannot capture the real pattern in data

Fix: Increase polynomial degree, add more features, reduce regularisation

Fitting a straight line to parabolic data

Optimal Fit (Just Right)

Train error:	Low
Test error:	Low
Degree:	Balanced

Cause: Model complexity perfectly matches the data structure

Fix: This is the goal — cross-validation helps find this point

Degree 2 polynomial for curved house price data

Overfitting (High Variance)

Train error:	Very low
Test error:	High
Degree:	Too high (deg 10+)

Cause: Model memorises training noise — fails completely on new data

Fix: Reduce degree, use Ridge/Lasso regularisation, get more data

Degree 15 polynomial fitting 10 training points exactly

Bias-Variance tradeoff: Too simple = high bias | Too complex = high variance | Use cross-validation to balance

Regression in the Real World — Input to Output

Real estate

Linear / Ridge

Inputs: House area (sq.ft), No. of rooms, Location, Floor

Predicted output → ₹52,00,000

Education

Polynomial

Inputs: Study hours/day, Attendance %, Previous marks, Sleep hrs

Predicted output → 78 / 100

Weather

SVR / Neural Net

Inputs: Humidity, Air pressure, Wind speed, Month

Predicted output → 142 mm rain

Healthcare

Linear / Lasso

Inputs: Age, Weight, Exercise hrs, Diet score

Predicted output → Blood pressure 120

Finance

DT Reg / Neural Net

Inputs: Trading volume, Prev. price, Market index, News score

Predicted output → Stock: ₹2,345

Agriculture

Ridge / SVR

Inputs: Rainfall, Temperature, Soil pH, Fertiliser qty

Predicted output → Yield: 42 kg

Key Assumptions & Best Practices in Regression

Linearity

Relationship between features and output is linear (for Linear Regression). Check with scatter plots.

Fix: Use Polynomial or SVR if relationship is curved

No multicollinearity

Input features should not be highly correlated with each other — can destabilise weights.

Fix: Use Ridge Regression or remove correlated features

Homoscedasticity

Error variance should be constant across all values of X — not increase or decrease with X.

Fix: Transform output (log, sqrt) or use weighted regression

Normal errors

Residuals (actual – predicted) should be normally distributed around zero.

Fix: Check with histogram of residuals; transform features if needed

Best practices

- Always scale features for SVR
- Check residual plots
- Use cross-validation
- Start with Linear → add complexity if needed

Summary — Regression

1. Regression = Supervised Learning task that predicts a CONTINUOUS NUMBER (quantity, price, score, measurement)
2. Linear Regression: best-fit straight line $\hat{Y} = mX + c$ (simple) or $\hat{Y} = w_1x_1+w_2x_2+...+b$ (multiple)
3. Polynomial Regression: curved line — adds X^2, X^3 terms. More flexible but higher overfitting risk
4. Ridge (L2): adds Σw_i^2 penalty — shrinks weights. Best for correlated features. Keeps ALL features
5. Lasso (L1): adds $\Sigma|w_i|$ penalty — can zero out weights. Automatic feature selection. Sparse model
6. SVR: fits a tube around data — only errors outside tube penalised. Handles non-linear and outlier data
7. DT Regression: tree splits using variance reduction → leaf predicts average value. Interpretable.
8. Metrics: MAE (average error), RMSE (penalises big errors), R^2 (% variance explained — closer to 1 = better)
9. Overfitting fix: Ridge/Lasso regularisation. Underfitting fix: higher degree polynomial or more features

Answer is a NUMBER → Regression | Start with Linear → Polynomial → Ridge/Lasso → SVR → Neural Net